

Dataset Field Explanations

NASA's full explanation can be found [here](#).

Dataset Columns

Identification Columns

- **kepid**: A unique ID number for the target star from the Kepler Input Catalog.
- **kepoi_name**: A tracking number for a specific potential planet (KOI) orbiting a star.

Exoplanet Archive Information

- **kepler_name**: The official public name given to a confirmed planet (e.g., Kepler-22b).
- **koi_disposition**: The final status of the object (e.g., Confirmed, Candidate, False Positive). *This is our target variable.*
- **koi_vet_stat**: Whether scientists are actively updating this data ("ACTIVE") or considered finalized ("DONE").
- **koi_vet_date**: The exact date when this object's parameters were last updated.

Project Disposition Columns

- **koi_pdisposition**: The Kepler pipeline's automated guess of what the object is, before human confirmation.
- **koi_score**: A confidence score from 0 to 1 on whether the pipeline thinks the object is actually a planet.
- **koi_fpflag_nt**: A flag (yes/no) indicating if the signal looks like a telescope glitch or strange star behavior rather than a planet.
- **koi_fpflag_ss**: A flag indicating if the signal is likely simply another star blocking the light (an eclipsing binary).
- **koi_fpflag_co**: A flag indicating if the light dip is actually coming from a different background star nearby.
- **koi_fpflag_ec**: A flag indicating if the signal is a duplicate or contamination from another known object.
- **koi_disp_prov**: The source or method used to determine the object's status.
- **koi_comment**: A text note explaining exactly why an object was marked as a false positive.

Transit Properties (The Physics of the Orbit)

- **koi_period**: How many days it takes for the planet to complete one full orbit around its star.
- **koi_time0bk / koi_time0**: The exact astronomical time when the very first transit was detected.
- **koi_eccen**: How oval-shaped the planet's orbit is.
- **koi_longp**: The angle of the planet's orbit at its closest point to the star.
- **koi_impact**: How close the planet passes to the exact center of the star's visual disk when it crosses in front of it.
- **koi_duration**: How many hours the transit event lasts from start to finish.
- **koi_ingress**: The time it takes for the planet to fully move onto the star's disk at the start of the transit.
- **koi_depth**: The percentage of the star's light that is blocked by the planet.
- **koi_ror**: The physical size of the planet compared directly to the physical size of its star.
- **koi_srho**: The estimated density of the star, calculated directly from the transit data.
- **koi_fittype**: The mathematical modeling method used to calculate the planet's characteristics.
- **koi_prad**: The estimated physical size of the planet, measured in Earth radii.
- **koi_sma**: The average distance between the planet and its star.
- **koi_incl**: The tilt angle of the planet's orbit relative to our line of sight from Earth.
- **koi_teq**: The estimated surface temperature of the planet based on its distance from the star.
- **koi_insol**: How much radiation the planet receives from its star, compared to what Earth gets from the Sun.
- **koi_dor**: The distance between the planet and star divided by the star's radius.
- **koi_limbdark_mod / koi_ldm_coeff**: Mathematical numbers used to account for the visual effect where a star looks darker at its outer edges.
- **koi_parm_prov**: Where the math for these specific transit parameters originally came from.

Threshold-Crossing Event (TCE) Information (The Data Signals)

- **koi_max_sngle_ev**: A statistical measure of how strong the single clearest transit signal was.
- **koi_max_mult_ev**: A statistical measure of the signal strength when looking at all the transits combined.
- **koi_model_snr**: Signal-to-noise ratio; how clear the transit signal is compared to background data.
- **koi_count**: The total number of potential planets found orbiting this specific star.
- **koi_num_transits**: How many times the planet was seen crossing the star in the dataset.

- **koi_tce_plnt_num / koi_tce_delivname**: Internal pipeline tracking numbers and delivery batch names.
- **koi_quarters**: Which specific multi-month observing periods (quarters) the telescope used to watch this star.
- **koi_trans_mod / koi_model_dof / koi_model_chisq**: Technical statistics on how well the mathematical models fit the observed data.
- **koi_datalink_dvr / koi_datalink_dvs**: Web links to the official NASA technical reports for this object.

Stellar Parameters (The Host Star)

- **koi_steff**: The surface temperature of the host star in Kelvin.
- **koi_slogg**: The strength of gravity at the star's surface.
- **koi_smet**: How much of the star is made up of heavy elements (like iron) compared to our Sun.
- **koi_srad**: The physical size of the star, compared to our Sun.
- **koi_smass**: The physical weight/mass of the star, compared to our Sun.
- **koi_sage**: How old the star is, measured in billions of years.
- **koi_sparprov**: Where the information about this star originally came from (e.g., ground-based observations or models).

KIC Parameters (Telescope Coordinates & Brightness)

- **ra / dec**: Right Ascension and Declination. These are the longitude and latitude coordinates for the star in the night sky.
- **koi_kepmag**: How bright the star appears through the Kepler telescope's main lens.
- **koi_gmag through koi_kmag**: How bright the star appears in various specific colors or wavelengths of light (like infrared or visible red/blue).

Pixel-Based KOI Vetting Statistics (Image Checks)

These fields together answer the following question: "Did the dip in light come from the exact center of our target star, or did it come from slightly off to the side, indicating it may be a background star?"

- **koi_fwm_sra / koi_fwm_sdec**: The calculated sky coordinates of where the transit signal came from.
- **koi_fwm_srao / koi_fwm_sdeco**: How much the center of the light shifted horizontally or vertically during the transit.
- **koi_fwm_prao / koi_fwm_pdeco**: How far the transit signal's location is from the target star's known location.

- **koi_fwm_stat_sig**: A percentage showing how confident the scientific community is that the transit happened directly on the target star.
- **koi_dicco_ and koi_dikco_ series**: Advanced image-difference checks measuring the exact visual distance (in arcseconds) between the normal image, the during-transit image, and the official catalog location.